Connecting Prometheus and Node Exporter.

What is Prometheus?

Prometheus is a **time-series monitoring system**.

It:

- Collects metrics
- Stores them as time-series data
- Allows querying using PromQL
- Triggers alerts

It works on a **pull model**.

What is Node Exporter?

Node Exporter is a lightweight agent.

It:

- Runs on a machine (Linux server)
- Collects OS-level metrics
- Exposes them via HTTP endpoint (/metrics)
- Default port: **9100**

It does NOT store data.
It does NOT analyze data.
It only exposes metrics.

we have to install the Prometheus on the master machine.

```
[root@ip-172-31-23-102 opt]# tar -xvf prometheus-2.52.0.linux-amd64.tar.gz
prometheus-2.52.0.linux-amd64/
prometheus-2.52.0.linux-amd64/LICENSE
prometheus-2.52.0.linux-amd64/promtool
prometheus-2.52.0.linux-amd64/console_libraries/
prometheus-2.52.0.linux-amd64/console_libraries/prom.lib
prometheus-2.52.0.linux-amd64/console_libraries/menu.lib
prometheus-2.52.0.linux-amd64/prometheus
prometheus-2.52.0.linux-amd64/consoles/
prometheus-2.52.0.linux-amd64/consoles/node-overview.html
prometheus-2.52.0.linux-amd64/consoles/prometheus.html
prometheus-2.52.0.linux-amd64/consoles/node.html
prometheus-2.52.0.linux-amd64/consoles/index.html.example
prometheus-2.52.0.linux-amd64/consoles/node-disk.html
prometheus-2.52.0.linux-amd64/consoles/node-cpu.html
prometheus-2.52.0.linux-amd64/consoles/prometheus-overview.html
prometheus-2.52.0.linux-amd64/prometheus.yml
prometheus-2.52.0.linux-amd64/NOTICE
[root@ip-172-31-23-102 opt]#
```

We have to install and run the node exporter in the slave machine.

```
[ec2-user@ip-172-31-23-90 opt]$ tar -xvf node_exporter-1.10.2.linux-amd64.tar.gz
node_exporter-1.10.2.linux-amd64/
node_exporter-1.10.2.linux-amd64/node_exporter
node_exporter-1.10.2.linux-amd64/LICENSE
node_exporter-1.10.2.linux-amd64/NOTICE
[ec2-user@ip-172-31-23-90 opt]$ cd cd
-bash: cd: cd: No such file or directory
[ec2-user@ip-172-31-23-90 opt]$
[ec2-user@ip-172-31-23-90 opt]$
[ec2-user@ip-172-31-23-90 opt]$ cd node_exporter-1.10.2.linux-amd64/
[ec2-user@ip-172-31-23-90 node_exporter-1.10.2.linux-amd64]$ ls
LICENSE  NOTICE  node_exporter
[ec2-user@ip-172-31-23-90 node_exporter-1.10.2.linux-amd64]$ ./node_exporter
time=2026-02-25T07:48:19.967Z level=INFO source=node_exporter.go:216 msg="Starting node_exporter" version="(version=1.10.2, branch=HEAD, revision=654f19dee6a0c41de78a8
d6d870e8c742cdb43b9)"
time=2026-02-25T07:48:19.967Z level=INFO source=node_exporter.go:217 msg="Build context" build_context="(go=go1.25.3, platform=linux/amd64, user=root@b29b4019149a, dat
```

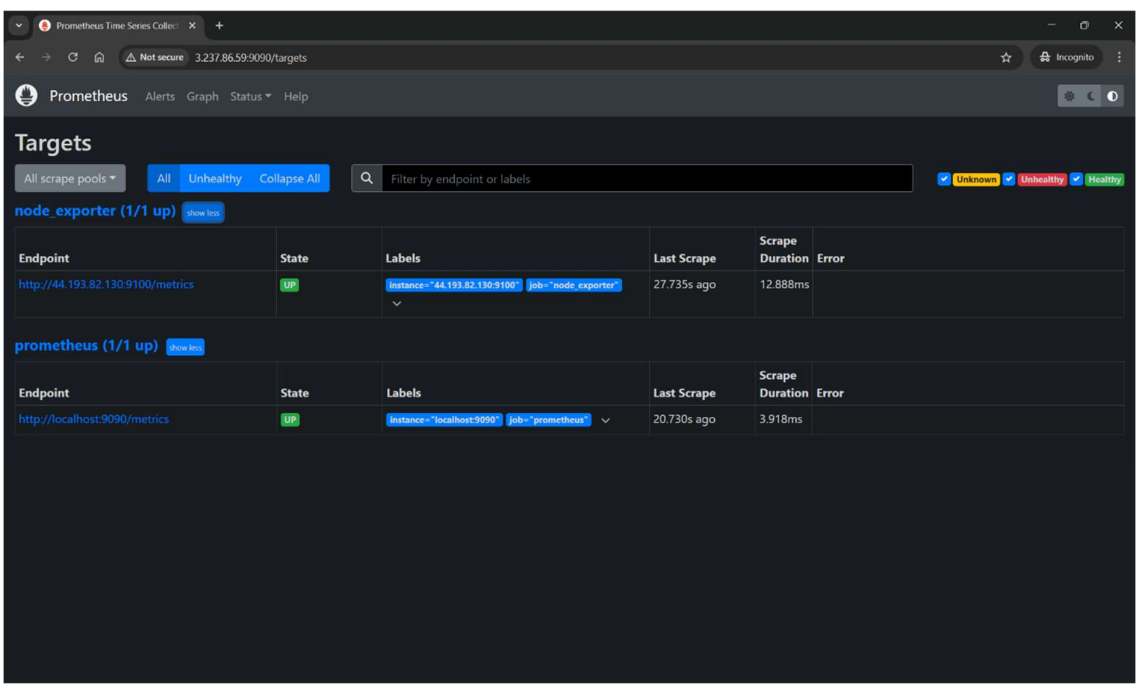
we have to connect to the master machine by using configure the Prometheus.yml file with the public ip of the slave machines.

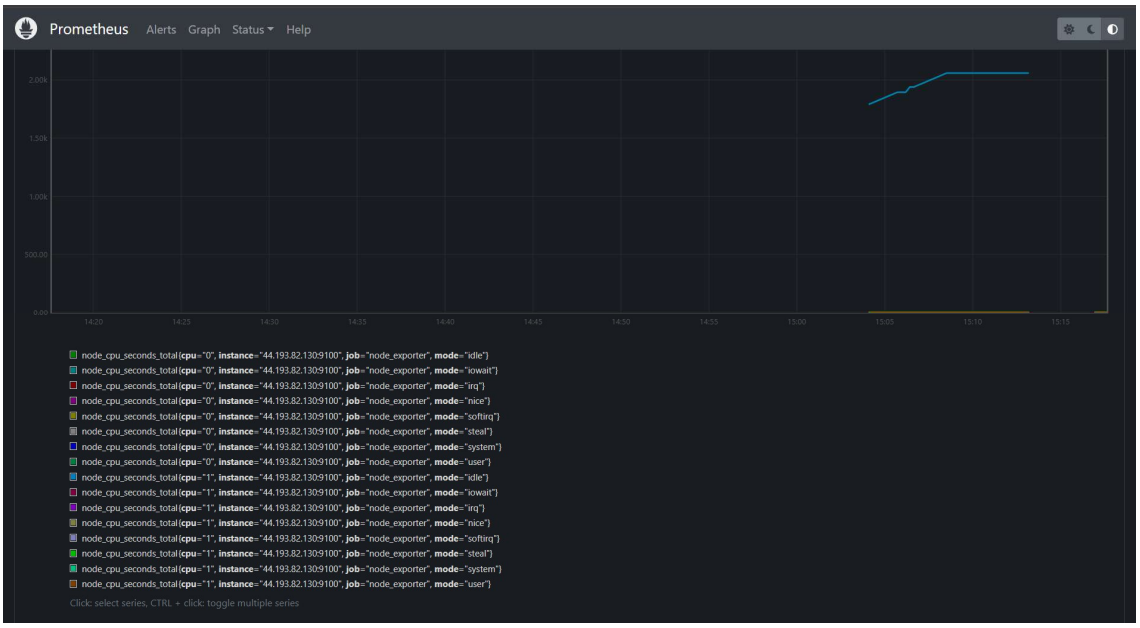
```
  GNU nano 8.3                                          prometheus.yml
alerting:
  alertmanagers:
    - static_configs:
        - targets:
          # - alertmanager:9093

# Load rules once and periodically evaluate them according to the global 'evaluation_interval'.
rule_files:
  # - "first_rules.yml"
  # - "second_rules.yml"

# A scrape configuration containing exactly one endpoint to scrape:
# Here it's Prometheus itself.
scrape_configs:
  # The job name is added as a label `job=<job_name>` to any timeseries scraped from this config.
  - job_name: "prometheus"

    # metrics_path defaults to '/metrics'
    # scheme defaults to 'http'.

    static_configs:
      - targets: ["localhost:9090"]
  - job_name: "node_exporter"

    # metrics_path defaults to '/metrics'
    # scheme defaults to 'http'.

    static_configs:
      - targets: ["44.193.82.130:9100"]

^G Help      ^O Write Out  ^F Where Is   ^K Cut       ^T Execute    ^C Location   M-U Undo      M-A Set Mark  M-] To Bracket  M-B Previous
^X Exit      ^R Read File  ^\ Replace    ^U Paste     ^J Justify    ^/ Go To Line M-E Redo      M-6 Copy      ^B Where Was    M-F Next
```

i-040a435ad7cabf433 (prometheus)

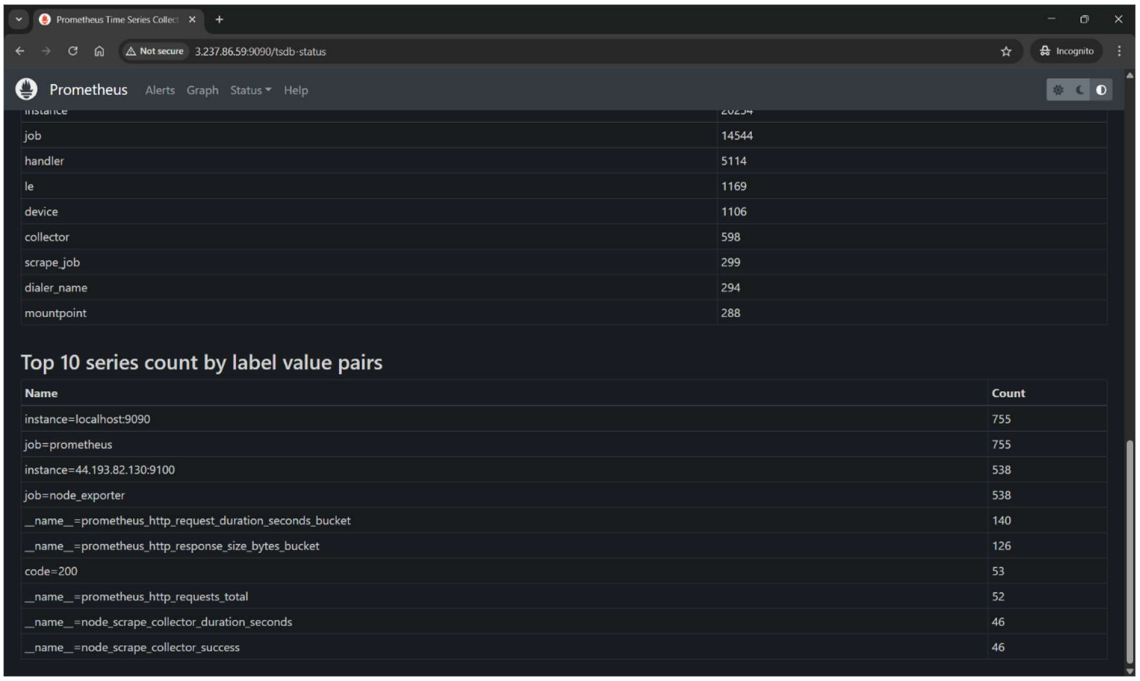
PublicIPs: 3.237.86.59 PrivateIPs: 172.31.13.222

Then we have connected to the slave machine.



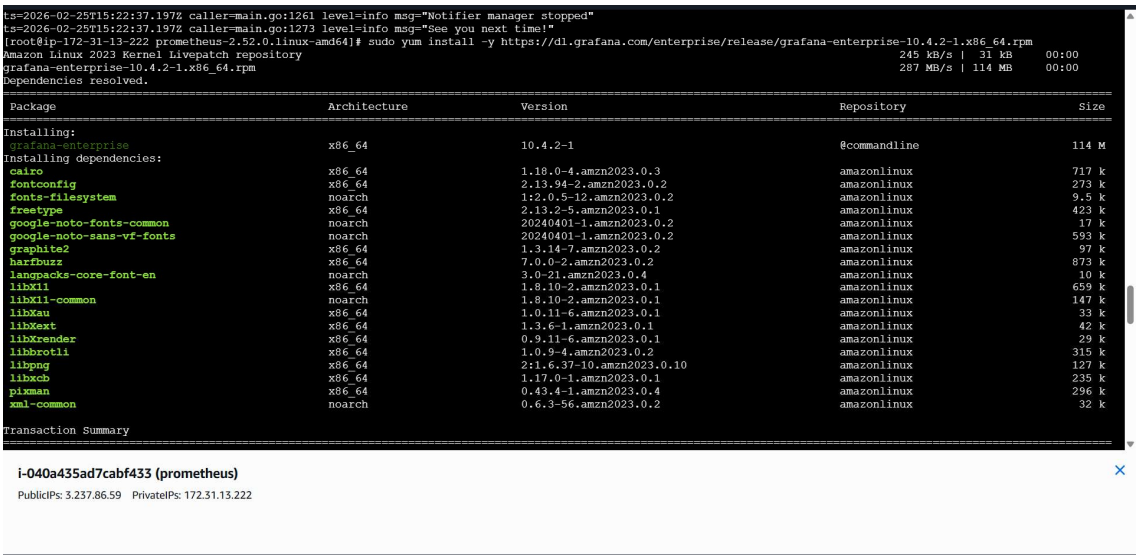

Prometheus
Alerts
Graph
Status
Help
2.00k
1.50k
1.00k
500.00
0.00
14:20
14:25
14:30
14:35
14:40
14:45
14:50
14:55
15:00
15:05
15:10
15:15
node_cpu_seconds_total{cpu="0", instance="44.193.82.130:9100", job="node_exporter", mode="idle"}
node_cpu_seconds_total{cpu="0", instance="44.193.82.130:9100", job="node_exporter", mode="iowait"}
node_cpu_seconds_total{cpu="0", instance="44.193.82.130:9100", job="node_exporter", mode="irq"}
node_cpu_seconds_total{cpu="0", instance="44.193.82.130:9100", job="node_exporter", mode="nice"}
node_cpu_seconds_total{cpu="0", instance="44.193.82.130:9100", job="node_exporter", mode="softirq"}
node_cpu_seconds_total{cpu="0", instance="44.193.82.130:9100", job="node_exporter", mode="steal"}
node_cpu_seconds_total{cpu="0", instance="44.193.82.130:9100", job="node_exporter", mode="system"}
node_cpu_seconds_total{cpu="0", instance="44.193.82.130:9100", job="node_exporter", mode="user"}
node_cpu_seconds_total{cpu="1", instance="44.193.82.130:9100", job="node_exporter", mode="idle"}
node_cpu_seconds_total{cpu="1", instance="44.193.82.130:9100", job="node_exporter", mode="iowait"}
node_cpu_seconds_total{cpu="1", instance="44.193.82.130:9100", job="node_exporter", mode="irq"}
node_cpu_seconds_total{cpu="1", instance="44.193.82.130:9100", job="node_exporter", mode="nice"}
node_cpu_seconds_total{cpu="1", instance="44.193.82.130:9100", job="node_exporter", mode="softirq"}
node_cpu_seconds_total{cpu="1", instance="44.193.82.130:9100", job="node_exporter", mode="steal"}
node_cpu_seconds_total{cpu="1", instance="44.193.82.130:9100", job="node_exporter", mode="system"}
node_cpu_seconds_total{cpu="1", instance="44.193.82.130:9100", job="node_exporter", mode="user"}
Click: select series, CTRL + click: toggle multiple series

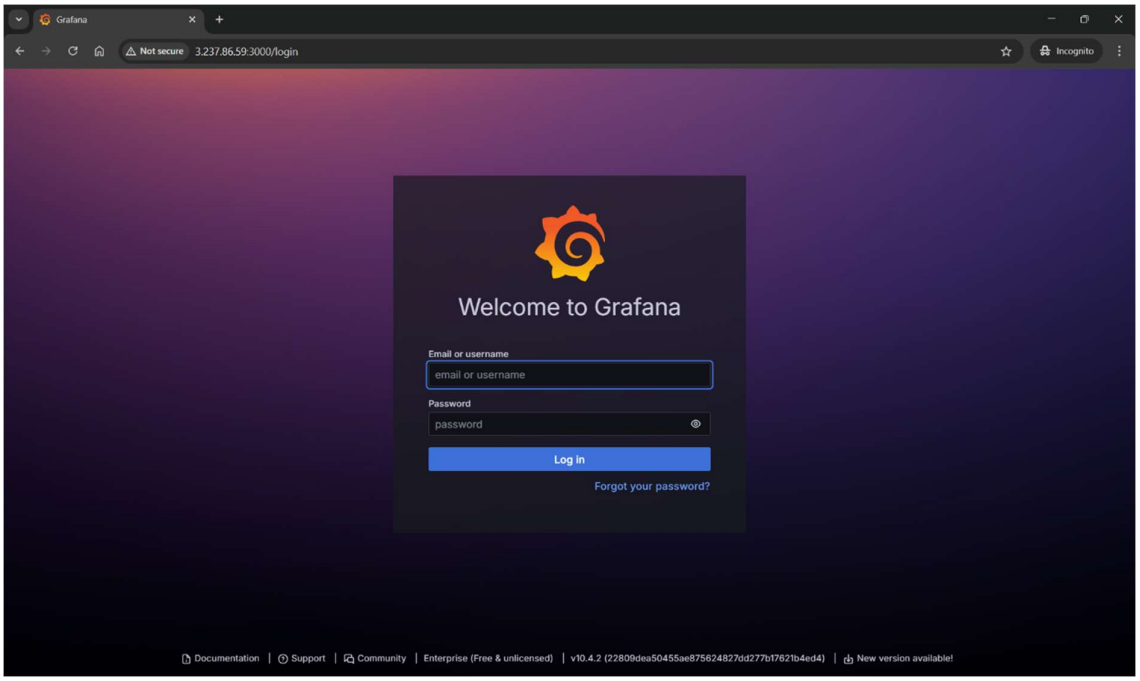
Prometheus Time Series Collect
Not secure
3.237.86.59:9090/graph?g0.expr=node_memory_Slab_bytes&g0.tab=0&g0.display_mode=lines&g0.show_exemplars=0&g0.range_input=1h
Incognito
Prometheus
Alerts
Graph
Status
Help
Table
Graph
Load time: 332ms
Resolution: 14s
Result series: 1
1h
End time
Res. (s)
Show Exemplars
55.05M
55.00M
54.95M
54.90M
54.85M
54.80M
14:20
14:25
14:30
14:35
14:40
14:45
14:50
14:55
15:00
15:05
15:10
15:15
node_memory_Slab_bytes{instance="44.193.82.130:9100", job="node_exporter"}
Remove Panel
Add Panel

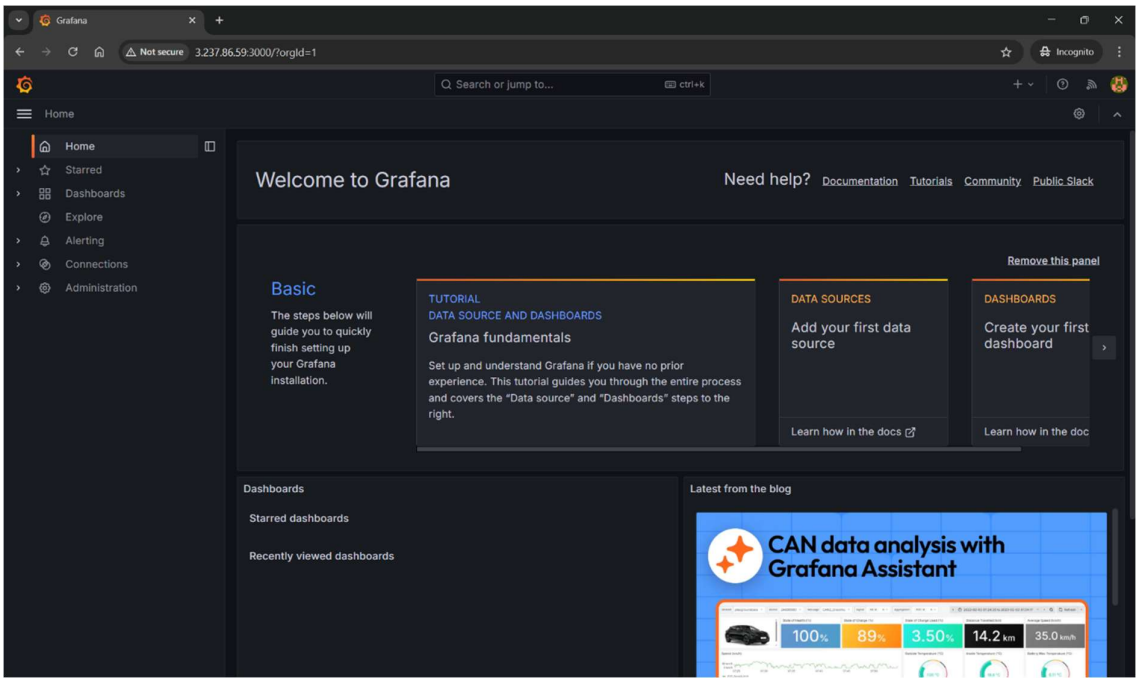


GRAFANA

Install the Grafana in master machine.




Grafana
Not secure
3.237.86.59:3000/login
Incognito
Welcome to Grafana
Email or username
email or username
Password
password
Log in
Forgot your password?
Documentation | Support | Community | Enterprise (Free & unlicensed) | v10.4.2 (22809dea50455ae875624827dd277b17621b4ed4) | New version available!


Grafana
Not secure
3.237.86.59:3000/?orgId=1
Incognito
Search or jump to...
ctrl+k
Home
Home
Starred
Dashboards
Explore
Alerting
Connections
Administration
Welcome to Grafana
Need help? Documentation Tutorials Community Public Slack
Remove this panel
Basic
The steps below will guide you to quickly finish setting up your Grafana installation.
TUTORIAL
DATA SOURCE AND DASHBOARDS
Grafana fundamentals
Set up and understand Grafana if you have no prior experience. This tutorial guides you through the entire process and covers the "Data source" and "Dashboards" steps to the right.
DATA SOURCES
Add your first data source
Learn how in the docs
DASHBOARDS
Create your first dashboard
Learn how in the doc
Dashboards
Starred dashboards
Recently viewed dashboards
Latest from the blog
CAN data analysis with Grafana Assistant
100%
89%
3.50%
14.2 km
35.0 km/h